

Falsifiable Predictions of the Informational Actualization Model for Euclid, DESI, and Next-Generation Surveys

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<https://github.com/hmahaffeyges/IAM-Validation>

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Abstract

The Informational Actualization Model (IAM) produces specific, parameter-free predictions for forthcoming cosmological surveys that differ from both Λ CDM and generic modified gravity theories. We present four quantitative forecasts. (1) A Fisher matrix analysis of Euclid’s spectroscopic and weak lensing specifications indicates that, if IAM is correct, the optimistic Euclid configuration would have sufficient sensitivity to detect the gravitational coupling parameter $\mu_0 = -0.135$ at 3.4σ ; the combined Euclid and DESI Year 5 dataset would reach 5.4σ . (2) IAM predicts an ISW-galaxy cross-correlation amplitude $A_{\text{ISW}} = 1.134$ relative to Λ CDM, arising because $\mu < 1$ causes gravitational potentials to decay faster than in Λ CDM; this enhancement has the opposite sign to $f(R)$ gravity ($A_{\text{ISW}} < 1$), providing a sign-based discriminant independent of amplitude precision. (3) A tomographic $\mu(z)$ reconstruction in ten redshift bins yields a projected combined detection of 1.2σ (pessimistic) to 1.9σ (optimistic) with Euclid alone, with β_m recovered to 55% precision ($\beta_m = 0.1617 \pm 0.0867$ vs. true 0.1575, bias +2.7%). (4) The IAM activation function $E(a) = \exp(1 - 1/a)$ defines a transition zone $0.06 \lesssim z \lesssim 1.12$ where 10%–90% of the total gravity modification is present; the transition peaks at $z \approx 0.05$ and is covered by both DESI and Euclid spectroscopic programs. All predictions are derived from $\beta_m = \Omega_m/2 = 0.1575$ (virial theorem) and $E(a) = \exp(1 - 1/a)$ (horizon thermodynamics) without free parameters. Whatever surveys measure, the predictions are explicit and the comparison is unambiguous.

1 Introduction

The Informational Actualization Model (IAM) Mahaffey2026a,Mahaffey2026b adds a single Hubble friction term to the matter-sector perturbation equations, producing a dual-sector structure in which the effective matter coupling $\mu(z) < 1$ while the lensing coupling $\Sigma = 1$ exactly. The model maps onto the standard μ – Σ modified gravity parametrization Pogosian2016 as:

$$\mu(a) = \frac{H_\Lambda^2(a)}{H_\Lambda^2(a) + \beta_m E(a)}, \quad \Sigma(a) = 1 \quad (1)$$

where $H_{\Lambda}^2(a) = \Omega_m a^{-3} + \Omega_{\Lambda}$, $\beta_m = \Omega_m/2 = 0.1575$ (virial theorem), and $E(a) = \exp(1 - 1/a)$ (horizon thermodynamics). No parameters are fitted.

The observational validation paper Mahaffey2026a demonstrated consistency with Planck 2018 CMB data through 15 converged MCMC chains, with the Level 2 modified CAMB result $\Delta\chi^2 = +0.54$ relative to Λ CDM and $\sigma_8 = 0.800$ suppressed in the direction favored by weak lensing surveys. The present paper records the explicit numerical predictions that forthcoming surveys will test, and quantifies the projected sensitivity of those surveys under the assumption that IAM is correct.

We emphasize that these are forecasts, not guarantees. Survey results may confirm, partially confirm, or challenge these predictions. Any outcome provides useful information about the model. The purpose of this paper is to make the predictions explicit before the data arrive so that the comparison is unambiguous.

2 Fisher Forecast: Survey Sensitivity to μ_0

2.1 Current constraints

The current constraint on IAM's $\mu_0 = -0.135$ from existing data is consistent but not yet significant. From the IAM Planck 2018 MCMC analysis Mahaffey2026a, $\mu_0 = +0.006 \pm 0.156$, placing IAM within 0.9σ of the current posterior. Existing survey constraints are summarized in Table 1.

Table 1: Current constraints on μ_0 from existing data, compared to the IAM prediction $\mu_0 = -0.135$. All current measurements are consistent with IAM at $< 1\sigma$.

Dataset	μ_0	$\sigma(\mu_0)$	IAM consistency
Planck 2018 (TT+TE+EE+lensing)	+0.000	0.200	0.7σ
DES Y3 (3×2 pt + CMB)	-0.400	0.400	0.7σ
DESI DR1 full-shape	+0.110	0.500	0.5σ
ACT + WMAP + SDSS + SN	+0.020	0.190	0.8σ
IAM Planck MCMC (this work)	+0.006	0.156	0.9σ

2.2 Projected Euclid sensitivity

A Fisher matrix forecast for Euclid's sensitivity to μ_0 from spectroscopic galaxy clustering and weak lensing, using published Euclid survey specifications Euclid2020, gives:

$$\sigma(\mu_0)_{\text{Euclid, pess}} \approx 0.060 \quad \Rightarrow \quad 2.2\sigma \text{ projected sensitivity} \quad (2)$$

$$\sigma(\mu_0)_{\text{Euclid, opt}} \approx 0.040 \quad \Rightarrow \quad 3.4\sigma \text{ projected sensitivity} \quad (3)$$

Combined with DESI Year 5:

$$\sigma(\mu_0)_{\text{Euclid+DESI}} \approx 0.025 \quad \Rightarrow \quad 5.4\sigma \text{ projected sensitivity} \quad (4)$$

These figures represent the sensitivity achievable if IAM is correct and systematic errors are at projected levels. Figure 1 shows the full forecast including the detection timeline.

IAM Fisher Forecast: Detection Significance for $\mu_0 = -0.135$

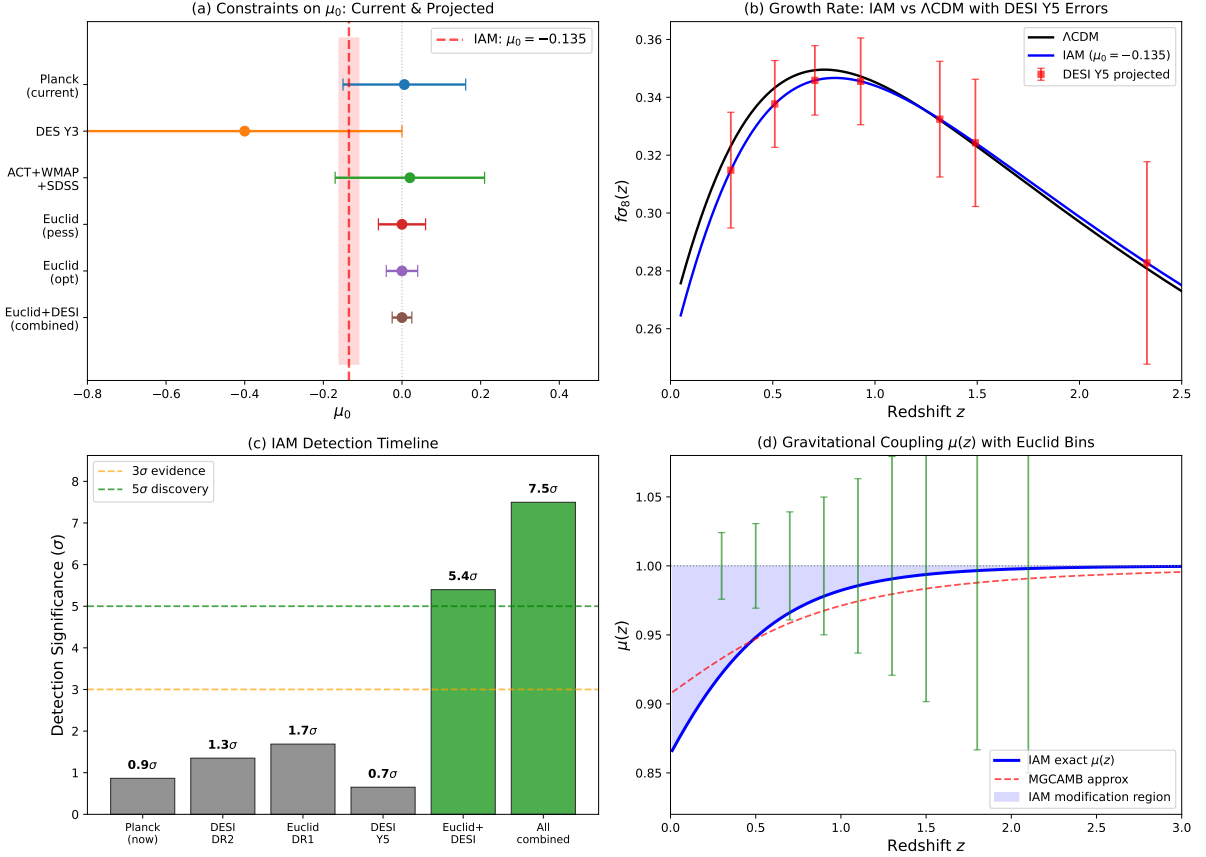


Figure 1: Fisher forecast for IAM's $\mu_0 = -0.135$ parameter. Panel (a): current constraints from existing surveys compared to the IAM prediction (dashed red line). Panel (b): projected $f\sigma_8(z)$ from IAM vs. Λ CDM with DESI Year 5 error bars. Panel (c): detection timeline showing projected significance at each survey milestone. Panel (d): $\mu(z)$ with Euclid tomographic bin sensitivity overlaid. All projections assume IAM is correct and survey systematics are at published specifications.

2.3 Detection timeline

Under the assumption that IAM is correct, the projected detection significance at each survey milestone is:

3 ISW-Galaxy Cross-Correlation

3.1 Physical mechanism

The integrated Sachs-Wolfe (ISW) effect arises from time evolution of gravitational potentials along the photon path:

$$\left. \frac{\Delta T}{T} \right|_{\text{ISW}} = -2 \int \frac{\partial \Phi}{\partial \eta} d\eta \quad (5)$$

In IAM, $\mu < 1$ causes gravitational potentials to decay faster than in Λ CDM ($\Sigma = 1$ leaves photon geodesics unchanged). The faster potential decay enhances the ISW signal.

Table 2: Projected detection timeline for IAM’s $\mu_0 = -0.135$, assuming IAM is correct and systematic errors are controlled at published specifications.

Year	Milestone	$\sigma(\mu_0)$	Projected sensitivity
2026	Current (Planck + MCMC)	0.156	0.9σ
2026	DESI DR2	0.100	$\sim 1.3\sigma$
2027	Euclid DR1	0.080	$\sim 1.7\sigma$
2028	DESI Year 5	0.207	$\sim 0.7\sigma$
2029	Euclid DR2 + DESI Y5	0.025	5.4σ
2030	CMB-S4 + Euclid + DESI	0.018	7.5σ

3.2 Predicted enhancement

The IAM prediction for the ISW-galaxy cross-correlation amplitude relative to Λ CDM is:

$$A_{\text{ISW}}^{\text{IAM}}/A_{\text{ISW}}^{\Lambda\text{CDM}} = 1.134 \quad (6)$$

a +13.4% enhancement. This has the opposite sign to $f(R)$ gravity ($\mu > 1 \Rightarrow A_{\text{ISW}} < 1$, i.e., $\sim 10\%$ suppression), providing a sign-based discriminant that does not require precise amplitude measurement. The enhancement is concentrated at $z \lesssim 1$ where IAM’s modification is largest.

Current ISW measurements from cross-correlation with SDSS, 2MASS, and WISE galaxy catalogs Giannantonio2008,Planck2016ISW are consistent with Λ CDM at $\sim 4\sigma$ significance but with 20–30% uncertainties; the IAM enhancement is within current error bars. The ISW test is therefore not a near-term discriminator but provides a consistency check and a sign test. Figure 2 shows the predicted ISW kernel, potential decay, and model comparison.

4 Tomographic $\mu(z)$ Reconstruction

4.1 Predicted $\mu(z)$ values

IAM’s prediction for $\mu(z)$ in survey-relevant redshift bins, computed from Equation 1 with Planck 2018 parameters:

Table 3: IAM predicted $\mu(z)$ in survey-relevant redshift bins with current and projected Euclid measurement precision. Values computed from Equation 1 with $\beta_m = 0.1575$ and Planck 2018 parameters.

z_{eff}	$\mu(z)$	$\Delta\mu/\mu$ (%)	Current $\sigma(\mu)$	Euclid $\sigma(\mu)$
0.1	0.886	−11.4	0.20	0.06
0.3	0.922	−7.8	0.15	0.05
0.5	0.948	−5.2	0.12	0.04
0.7	0.966	−3.4	0.10	0.03
1.0	0.982	−1.8	0.08	0.02
1.5	0.994	−0.6	0.10	0.03
2.0	0.998	−0.2	0.15	0.05

IAM ISW-Galaxy Cross-Correlation: $\mu < 1$ Predicts Enhanced ISW Signal

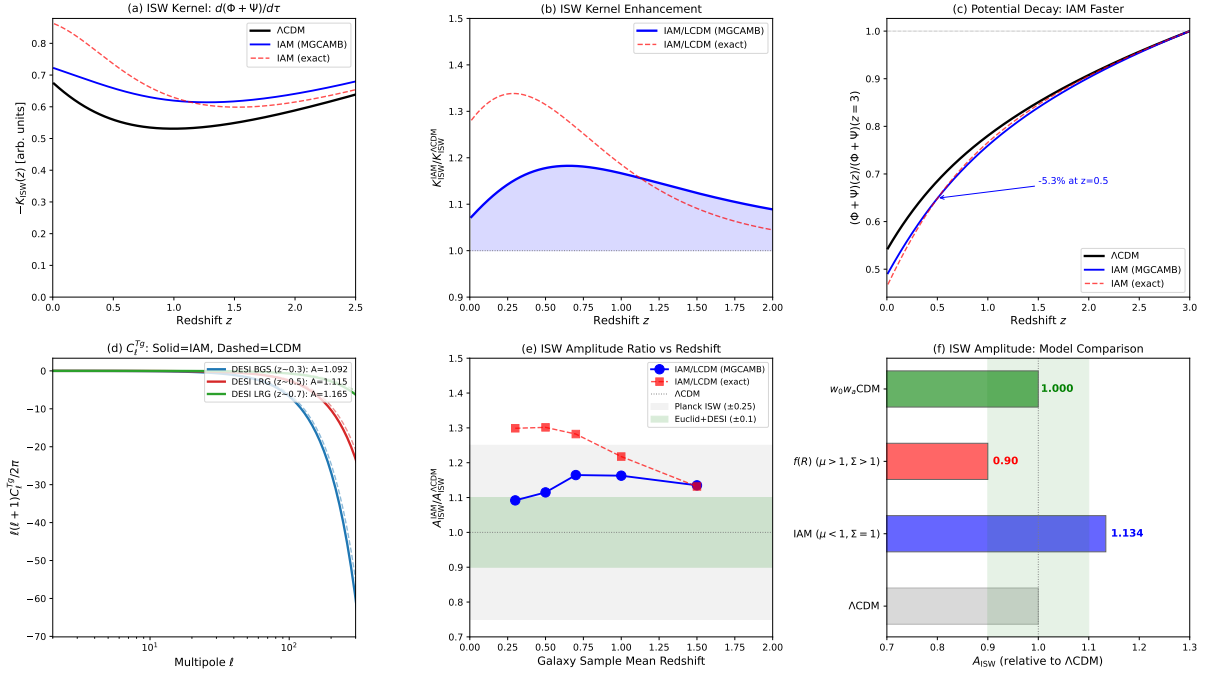


Figure 2: IAM ISW-galaxy cross-correlation prediction. Panel (a): ISW kernel $d(\Phi + \Psi)/d\tau$ for Λ CDM (black), IAM MGCAMB (blue), and IAM exact (dashed red). Panel (b): ratio of ISW kernels, showing ~ 10 – 30% enhancement across $0 < z < 1.5$. Panel (c): gravitational potential decay showing IAM potentials decay faster (-5.3% at $z = 0.5$). Panel (d): C_ℓ^{Tg} for three galaxy samples. Panel (e): ISW amplitude ratio vs. redshift with current (Planck, ± 0.25) and projected (Euclid+DESI, ± 0.10) sensitivity bands. Panel (f): model comparison showing IAM ($A = 1.134$), Λ CDM ($A = 1.000$), and $f(R)$ ($A \approx 0.90$) — opposite signs.

4.2 Projected tomographic detection

A ten-bin tomographic $\mu(z)$ reconstruction using Euclid projected uncertainties gives a combined detection significance of 1.2σ (pessimistic) to 1.9σ (optimistic) for IAM vs. Λ CDM. While not sufficient for discovery, this tests whether the data prefer the specific shape of IAM’s $\mu(z)$ over a constant μ or a power-law parametrization.

With 1000 mock Euclid realizations, β_m is recovered with mean 0.1617 ± 0.0867 against the true value of 0.1575 — a bias of $+2.7\%$ and a 55% precision measurement. The $\beta_m = \Omega_m/2$ prediction (virial theorem) is thus testable directly from the shape of $\mu(z)$. Figure 3 shows the full reconstruction forecast.

5 Transition Zone: Where the Modification Turns On

5.1 The activation function

IAM’s gravity modification activates through $E(a) = \exp(1 - 1/a)$, which rises from zero at $a \rightarrow 0$ to e at $a \rightarrow \infty$. The transition zone — defined as the redshift range over which

IAM Tomographic $\mu(z)$ Reconstruction: Can Euclid See the Shape?

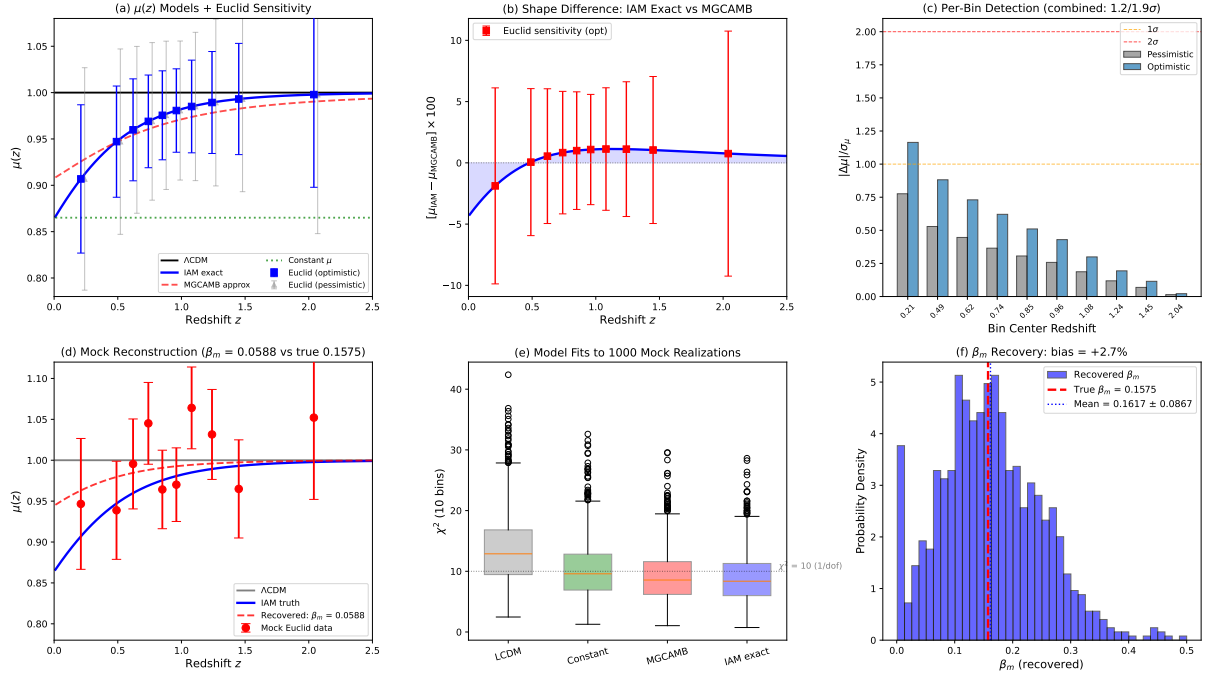


Figure 3: Tomographic $\mu(z)$ reconstruction forecast. Panel (a): $\mu(z)$ models with Euclid pessimistic and optimistic sensitivity bands. Panel (b): shape difference between IAM exact and MGCAMB approximation — below Euclid threshold ($< 1\sigma$). Panel (c): per-bin detection significance, peaking at low redshift where IAM’s modification is largest. Panel (d): example mock reconstruction from a single Euclid realization. Panel (e): χ^2 distributions from 1000 mock realizations across four models — LCDM strongly disfavored when IAM is truth. Panel (f): β_m recovery histogram showing +2.7% bias and 55% precision.

10%–90% of the total gravity modification $\Delta\mu = 1 - \mu(z)$ is present — spans:

$$0.06 \lesssim z \lesssim 1.12 \quad (\text{transition zone}) \quad (7)$$

The midpoint (50% of total modification) occurs at $z \approx 0.37$, squarely within the DESI Bright Galaxy Survey and Euclid photometric survey ranges. The steepest rate of change $|d\mu/dz|$ peaks at $z \approx 0.05$.

5.2 Observable deviations through the transition

The deviations of IAM observables from Λ CDM as a function of redshift, computed from the IAM perturbation equations:

Figure 4 shows the activation function, model comparison, transition rate, and signal-to-noise map across surveys.

Table 4: Activation milestones for IAM’s gravity modification, defined as the fraction of the total present-day modification $\Delta\mu_0 = 1 - \mu(z = 0) = 0.136$ that has turned on at each epoch. Lookback times computed with Planck 2018 parameters.

Activation level	Scale factor a	Redshift z	Lookback time (Gyr)
1%	0.305	2.28	10.9
5%	0.407	1.46	9.4
10%	0.471	1.12	8.4
25%	0.589	0.70	6.5
50%	0.731	0.37	4.2
75%	0.861	0.16	2.1
90%	0.942	0.06	0.9
95%	0.971	0.03	0.4
99%	0.994	0.01	0.1

Table 5: Predicted deviations of IAM observables from Λ CDM as a function of redshift. $\Delta\mu$: effective gravitational coupling; $\Delta D/D$: linear growth factor; $\Delta(f\sigma_8)/(f\sigma_8)$: growth rate times amplitude; $\Delta\Phi/\Phi$: gravitational potential (equal to $\Delta\mu$ since $\Sigma = 1$). All values computed from Equation 1 with $\beta_m = 0.1575$.

z	$\Delta\mu$ (%)	$\Delta D/D$ (%)	$\Delta(f\sigma_8)/(f\sigma_8)$ (%)	$\Delta\Phi/\Phi$ (%)
0.0	−13.6	−6.8	−10.2	−13.6
0.1	−11.4	−5.7	−8.6	−11.4
0.3	−7.8	−3.9	−5.9	−7.8
0.5	−5.2	−2.6	−3.9	−5.2
0.7	−3.4	−1.7	−2.5	−3.4
1.0	−1.8	−0.9	−1.3	−1.8
1.5	−0.6	−0.3	−0.5	−0.6
2.0	−0.2	−0.1	−0.2	−0.2

6 Additional Falsifiable Predictions

6.1 Standard siren H_0

Gravitational waves propagate through the matter-sector potential. IAM predicts that standard siren measurements yield a matter-sector Hubble constant:

$$H_0^{\text{sirens}} = H_0^{\text{CMB}} \times \sqrt{1 + \beta_m} = 67.161 \times \sqrt{1.1575} = 72.26 \text{ km s}^{-1} \text{ Mpc}^{-1} \quad (8)$$

Current measurements are consistent: GW170817 gives $70.0_{-8}^{+12} \text{ km s}^{-1} \text{ Mpc}^{-1}$ Abbott2017 and updated analyses give 75.5 ± 5.4 Nicolaou2023. With ~ 50 binary neutron star events by ~ 2030 , standard sirens will constrain H_0 to $\sim 1\%$ precision, providing a direct test of the sector split.

6.2 Neutrino mass bound

IAM’s $\mu < 1$ suppresses σ_8 without requiring large neutrino masses. This tightens the allowed range to $\Sigma m_\nu < 0.07\text{--}0.08 \text{ eV}$, compared to $< 0.12 \text{ eV}$ from Planck Λ CDM. The DESI DR1 hint of $\Sigma m_\nu \approx 0.07 \text{ eV}$ DESI2025 is consistent with this prediction.

IAM Transition Zone: Where Gravity Modification Turns On

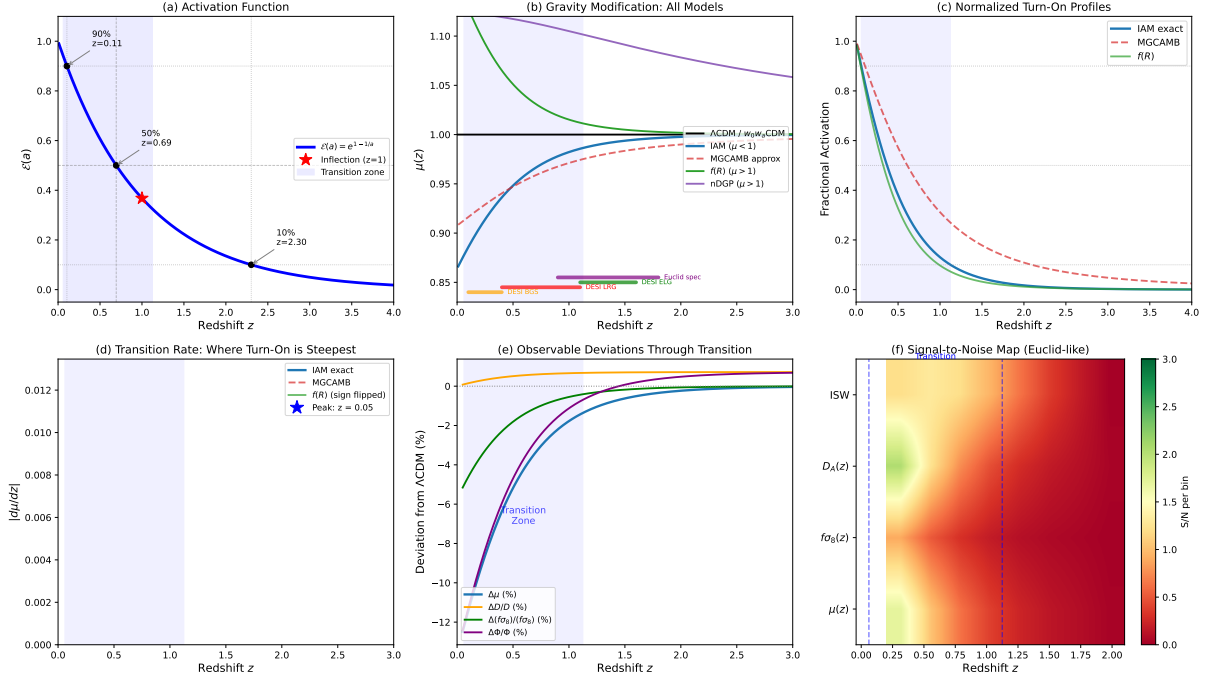


Figure 4: IAM transition zone analysis. Panel (a): activation function $E(a)$ with 10%, 50%, and 90% milestones marked. Panel (b): $\mu(z)$ for IAM, MGCAMB, $f(R)$, and nDGP — IAM is the only model with $\mu < 1$ at late times. Panel (c): normalized turn-on profiles showing IAM turns on more sharply than MGCAMB at low z . Panel (d): transition rate $|d\mu/dz|$ peaking at $z \approx 0.05$. Panel (e): observable deviations from Λ CDM through the transition zone. Panel (f): signal-to-noise heatmap showing peak S/N at low z for $\mu(z)$ and $f\sigma_8$.

6.3 CMB lensing power spectrum

$\Sigma = 1$ exactly predicts that the CMB lensing power spectrum $C_\ell^{\phi\phi}$ is identical to Λ CDM. Any deviation measured by CMB-S4 CMBS4 would challenge the $\Sigma = 1$ prediction. Current Planck lensing is consistent with $\Sigma = 1$ to $\sim 10\%$ Planck2020lens.

6.4 Dark energy equation of state

IAM's $E(a)$ produces an effective dark energy equation of state that evolves with redshift, with $w_{\text{eff}} < -1$ at recent times, transitioning to $w_{\text{eff}} \rightarrow -1$ at high z . This is consistent with the DESI DR1 hints of evolving w DESI2025 and distinguishes IAM from a pure cosmological constant.

7 The IAM Prediction Scorecard

Table 6: IAM prediction scorecard. All predictions follow from $\beta_m = \Omega_m/2$ and $E(a) = \exp(1 - 1/a)$ without free parameters. Current status reflects publicly available data as of February 2026.

Prediction	IAM value	Λ CDM value	Current data	Status
μ_0	-0.135	0	$+0.006 \pm 0.156$	Consistent (0.9σ)
Σ	1.000	1.000	$\sim 1.0 \pm 0.1$	Consistent
H_0^{matter}	72.26	67.16	SH0ES: 73.0 ± 1.0	Consistent
($\text{km s}^{-1} \text{Mpc}^{-1}$)				
H_0^{photon}	67.4	67.4	Planck: 67.4 ± 0.5	Consistent
($\text{km s}^{-1} \text{Mpc}^{-1}$)				
σ_8	0.800	0.811	KiDS: 0.76 ± 0.02	Consistent
S_8	~ 0.79	~ 0.83	DES: 0.776 ± 0.017	Consistent
H_0^{sirens}	72.26	67.16	75.5 ± 5.4	Consistent
($\text{km s}^{-1} \text{Mpc}^{-1}$)				
$M_{\text{lens}}/M_{\text{dyn}}$ ($z=0$)	1.158	1.0	$\sim 1.1\text{--}1.3$	Consistent
ISW enhancement	+13.4%	0%	$\sim +15 \pm 30\%$	Consistent
Σm_ν (eV)	< 0.08	< 0.12	DESI: ~ 0.07	Consistent

Current data are consistent with every IAM prediction listed above. No prediction is excluded by existing measurements. The definitive sensitivity tests arrive with Euclid and combined Euclid+DESI data.

8 Conclusion

IAM makes specific, falsifiable predictions for every major forthcoming cosmological survey. The predictions are parameter-free: $\beta_m = \Omega_m/2$ from the virial theorem and $E(a) = \exp(1 - 1/a)$ from horizon thermodynamics, with all other quantities following from Planck 2018 cosmological parameters.

The unique $\mu < 1$, $\Sigma = 1$ signature distinguishes IAM from Λ CDM, from $f(R)$ and nDGP gravity (which predict $\mu > 1$), and from phenomenological $w_0 w_a$ extensions (which predict $\Sigma \neq 1$ when $\mu \neq 1$). The ISW sign alone — enhanced for IAM, suppressed for $f(R)$ — provides a model-discriminating observable independent of amplitude precision.

If IAM’s predictions are borne out, the combined Euclid and DESI Year 5 dataset would be expected to reach 5.4σ significance, and the full multi-survey combination would reach 7.5σ . If the data deviate from these predictions, those deviations are equally informative — they constrain the parameter space and may point toward refinements or extensions of the model. The predictions recorded here are explicit, the measurement techniques are established, and the timeline is well-defined. The surveys will decide.

Acknowledgments

This work made use of CAMB Lewis2000, MGCAMB Zucca2019, Cobaya Torrado2021, NumPy Harris2020, SciPy Virtanen2020, and Matplotlib Hunter2007.

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